

PARSHURAM DESHPANDE

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OBJECTIVE

Electronics and Telecommunication Engineering student with hands-on experience in CPU architecture, RTL design, design verification, and open-source EDA workflows. Passionate about processor design, RTL-to-GDS implementation, and CAD automation, with experience in developing hardware design workflows and semiconductor automation tools.

PROFESSIONAL EXPERIENCE

CAD Developer Intern

July 2026 – Present

Stealth Semiconductor Startup

Remote

- Developing CAD workflows by integrating open-source EDA tools into a unified design environment for analog and digital IC design.
- Evaluating RTL-to-GDS workflows to identify automation opportunities and improve engineering productivity.
- Researching and benchmarking open-source EDA tools and simulators while contributing to workflow design and technical documentation.

Intern – Analog Design Automation

April 2026 – June 2026

UST

Remote

- Contributed to an automated analog amplifier design workflow integrating parameter prediction, SPICE netlist generation, and simulation optimization.
- Worked with Cadence Virtuoso, NgSpice, and Python for scripted analog design workflow automation and performance analysis.
- Gained exposure to design parameter sweeps and simulation result analysis, building a foundation for performance modeling methodologies.

PROJECTS

MiniRISC – 8-bit Single-Cycle Processor Design | *SystemVerilog, Computer Architecture, RTL Design*

- Designed and verified an 8-bit single-cycle RISC processor from scratch, implementing the complete datapath including Program Counter, Register File, ALU, Control Unit, Instruction/Data Memory, and Write-Back logic.
- Developed a parameterized, stage-based RTL architecture (Fetch, Decode, Execute, Memory, Write-Back) to support future migration to a pipelined processor.
- Built self-checking SystemVerilog testbenches to verify ALU operations, control logic, register file, instruction decoding, immediate generation, memory access, and write-back functionality using simulation and waveform analysis.

FPGA-Based Pedestrian Detection Exploration | *Verilog HDL, Lattice CertusPro-NX FPGA*

- Implemented and tested camera interfacing hardware and RTL post-processing modules on the Lattice CertusPro-NX FPGA platform.
- Performed hardware integration and validation as part of the Lattice Design Contest while evaluating FPGA-based edge AI deployment workflows.

TECHNICAL SKILLS

Programming & HDL: C, Python, Verilog HDL, SystemVerilog, Linux/Shell Scripting

Verification: SystemVerilog Testbench Development, RTL Simulation, Waveform Debugging, Functional Verification

EDA Tools: ModelSim, Cadence Virtuoso (ADE L, ADE XL, Layout XL), Cadence Genus, NgSpice, Quartus, Xilinx ISE 14.7

FPGA Platforms: Lattice CertusPro-NX, Spartan-6

Areas of Interest: CPU Microarchitecture, RTL Design, Design Verification, ASIC Design Flow, Physical Design, Open-Source EDA

EDUCATION

Vishwakarma Institute of Technology

Pune, India

Bachelor of Engineering, Electronics and Telecommunication Engineering

2023 – 2027

* CGPA: 8.91 / 10

* Relevant Coursework: Computer Architecture, Digital Electronics, VLSI Design, CMOS Analog Design